MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Saihajvir Singh

B. Tech Computer Engineering (2023-27)
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Summary

Computer Engineering undergraduate with hands-on experience in Data Engineering, Business Intelligence, and Software Development. Proficient in Python, SQL, Power BI, DAX, Java, React.js, and Node.js. Built machine learning models and full-stack applications, and contributed to wealth management analytics during an internship at HDFC Bank.

Education

MPSTME, NMIMS University — B.Tech in Computer Engineering 2023 - 2027
CGPA: 3.09/4.00
DAV Public School — HSC: 77.00 | CBSE: 86.00 2020 - 2023

Work Experience

HDFC Bank – Wealth Management Intern June 2026 – Aug 2026

- Developed interactive Power BI dashboards for AUM analysis, client segmentation, and investment recommendation tracking.
- Created DAX measures and automated ETL/data transformation workflows using Power Query and Microsoft Excel.
- Analysed wealth management portfolios to generate insights on holdings, recommendations, and investment performance.

Projects

PaisaPilot – AI Personal Finance Management System 2026
Technologies: *React Native, FastAPI, PostgreSQL, scikit-learn*

- Built an AI finance app for Indian youth: transaction categorisation, forecasting and anomaly detection.
- Merchant-grouped evaluation (75.5% accuracy, AUC 0.918); paper accepted at ICCTRDA 2026.

TRACE-X – Explainable Digital Forensics Engine 2026
Technologies: *Python, FastAPI, SQLAlchemy, PostgreSQL, React*

- Built a forensics engine reconstructing event timelines and correlation graphs with explainable findings.
- Evaluated on 12 scenarios (98 events) with an automated test suite and a full project report.

RootLens – Serverless Incident Detection (AWS) 2026
Technologies: *Python, AWS Lambda, CloudWatch, DynamoDB, SNS, Gemini API*

- Built a serverless pipeline: CloudWatch alarms, Gemini-based diagnosis, DynamoDB storage, SNS alerts.
- 10 fault-injection trials: 5/7 correct diagnoses (71.4%), no false alarms on normal runs.

Wildfire Detection System (Deep Learning) 2025
Technologies: *Python, TensorFlow, OpenCV, CNN*

- Built a CNN model using 4-channel (RGB + Thermal) imagery for wildfire detection, achieving 95% accuracy.
- Published as a research paper at SmartCom Conference, Pune (2025).

AI BlackJack Game 2025
Technologies: *Java, Python, AI, Probability Algorithms*

- Developed an AI-based Blackjack game using probability-based decision-making and rule-based game logic.
- Designed the game engine, scoring, card dealing, and AI strategy for realistic gameplay.

Certifications & Publications

Python Basics & Python Data Structures – University of Michigan (Coursera) | Managing Projects using AI (Coursera) | Ethical Hacker – Cisco Networking Academy | Full Stack Development – Meta (Coursera) | AWS Academy Graduate - Cloud Architecting

Core Skills

- **Programming:** Python, Java, C, C++, JavaScript, SQL
- **Full Stack:** HTML5, CSS3, React.js, Next.js, Node.js, Express.js, REST APIs, Tailwind CSS
- **Databases & AI/ML:** MongoDB, Firebase, Supabase | TensorFlow, OpenCV, Deep Learning, CNN
- **Data Analytics & BI:** Power BI, DAX, Power Query, Microsoft Excel, Data Visualization
- **Core CS & Tools:** DSA, OOP, DBMS, OS, Computer Networks | Git, GitHub, VS Code, Android Studio

Soft Skills: Leadership, Problem Solving, Analytical Thinking, Team Collaboration, Communication, Adaptability.

Achievements and Positions of Responsibility

- Finalist – LyondellBasell Hackathon
- Head of Photography Department – IET MPSTME (2024–25)
- Senior Executive – Outreach Department, Student Council (Sattva) (2025–26)
- Sub Head – Outreach Department, The Editorial Project (2025–26)
- Participant – Smart India Hackathon (SIH) (2025)
- Volunteer – Hunjan Hospital (2024)